

Section 7.4: Control Devices (Examples)

Computers can be used to control a range of devices through the use of sensors. Some of these devices include:

- **Security Lights**
- **Burglar Alarms**
- **Central Heating Controllers**
- **Computer Controlled Greenhouses**
- **Automatic Cookers**
- **Automatic Washing Machines**
- **Microwave Ovens**

Key Concepts of this section:

- # Understand how computers can **control devices automatically**.
- # Know why computers are **more effective at controlling devices** than people.
- # Be able to discuss different **examples of devices** that can be controlled by computers.

Controlling stuff using Computers

The role of the sensor

- # In section 7.3 we looked at how sensors could be used alongside computers to measure quantities (like temperature, light intensity etc) and then log changes in these quantities.

Sensors can also be used alongside computers to **control different devices**. The process goes like this:

Input - Sensor detects data in the environment around it

Process - Data passed to a computer (microprocessor) inside the device which analyses it and decides what action to take. The computer sends instructions to the device telling it what to do.

Output - The device would carry out the instructions.



Key Words:

Sensor, Input, Process, Output.

Examples of Sensors

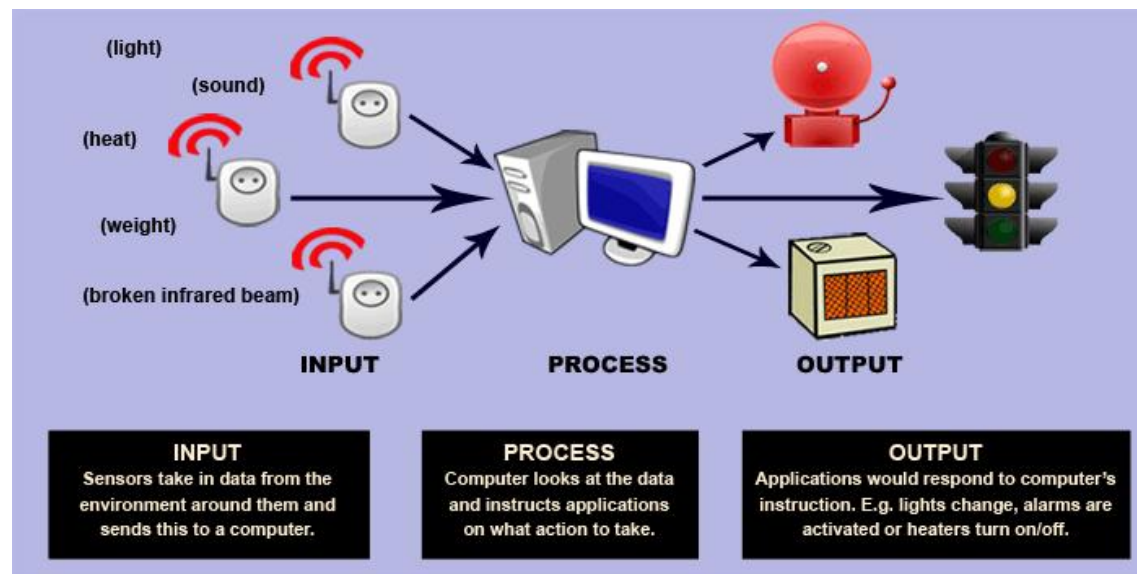
Temperature/Heat - Detect changes in temperature.

Light Sensor - Used to detect the brightness of light.

Sound Sensor - Detect the loudness of sound.

Humidity Sensor - Detect water in air.

Infra-red Sensor - Detect movement (Used in burglar alarms).

Picture Example:

Load/Weight Sensor - Detect weight (Used in digital scales).

Proximity Sensor - Detect how near/far something is.

REMEMBER!

Most **sensors are Analogue** so we can only connect them to a computer using an **Analogue to Digital Converter**.

Advantages of using a computer to control devices rather than people

Computers have certain **advantages over people** when it comes to controlling devices. For example:

- Cheaper -** ➔ If a computer is monitoring and controlling applications, you **do not need to employ people**.
- Higher Work Rate -** ➔ Computers can **control applications all day, every day** without getting tired or bored.
- Safer -** ➔ Computers can **work in conditions** that would be **too dangerous for people**. Examples include chemical plants, radioactive sites and extremely cold areas (antarctic).
- Accuracy -** ➔ Computers will respond to inputs from sensors **accurately every time**. E.g. a heater will be switched on as soon as the temperature falls below 10°C.
- Speed -** ➔ Computers will respond to data received from sensors **very quickly**. E.g. as soon as an infrared sensor detects an intruder the alarm will sound.



Radioactive sites can be too dangerous for people to work in.

Examples of Devices that are controlled by a Computer



Key Words:

Security Light, Alarm System, Automatic Heating System, Automatic Greenhouse.

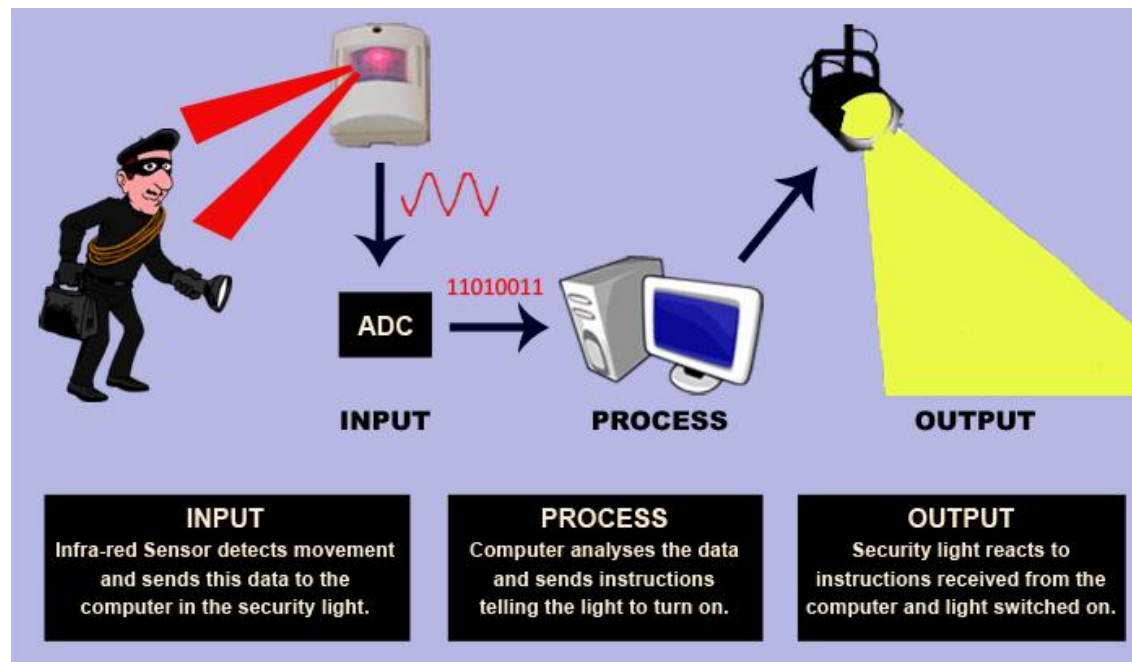
The following examples discuss a number of devices that can be controlled by a computer:

1 Security Lights

- # Security lights use **infra-red sensors** to **detect movement**.
- # Movement is detected when something (man or beast) **breaks the infra-red beam**. When this happens, the system will **turn on the light**.
- # If nothing else breaks the beam after a period of time, the system will turn the light off.



Security lights with built-in infra-red sensor



Video of Infra-Red Security Light

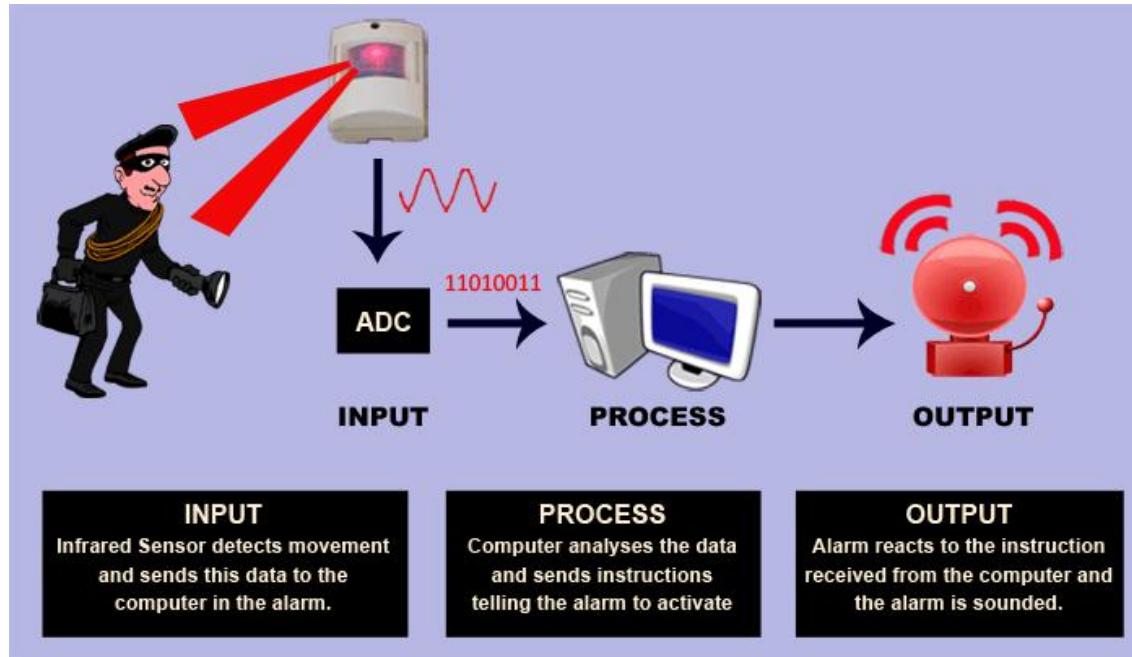
2 Burglar Alarms

Burglar alarms work in almost the same way as a security light system.

Input - Infra-red sensor detects movement when the sensor is broken and this information is sent to the computer

Process - Computer makes the decision to sound the alarm

Output - Alarm is activated.



A modern alarm system with infra-red sensors

3 Central Heating

Modern central heating systems can be programmed to **maintain a constant desirable temperature.**

Let's imagine our perfect room temperature was set at **25°C** - we could use sensors and a computer to:

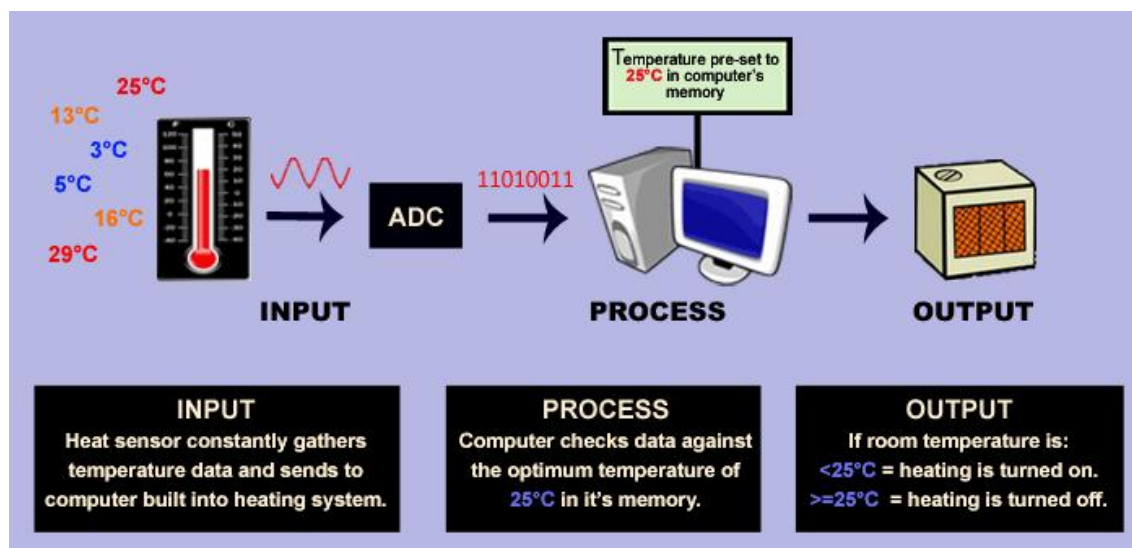
- **Maintain the optimum temperature of 25°C**
- **Turn heating off when it rises above 25°C**
- **Turn heating on when it falls below 25°C**

Here is how it would work:

Input - Heat sensor detects the current temperature and sends this data to the computer built into the heating system

Process - Computer would check the sensor heat data against the temperature setting stored in it's memory. Computer decides if the heating needs to be turned on (or off)

Output - Heating is switched on or off.



Programmable heating system

4 Computer Controlled Greenhouse

Automatic greenhouses can provide the **optimum conditions for growing plants** by using computer control.

Computers can be used to **monitor conditions** and **control a range of applications** (devices) to keep the **perfect conditions constant**.

Sensors needed to collect data:

- Light sensor
- Moisture sensor
- Heat sensor

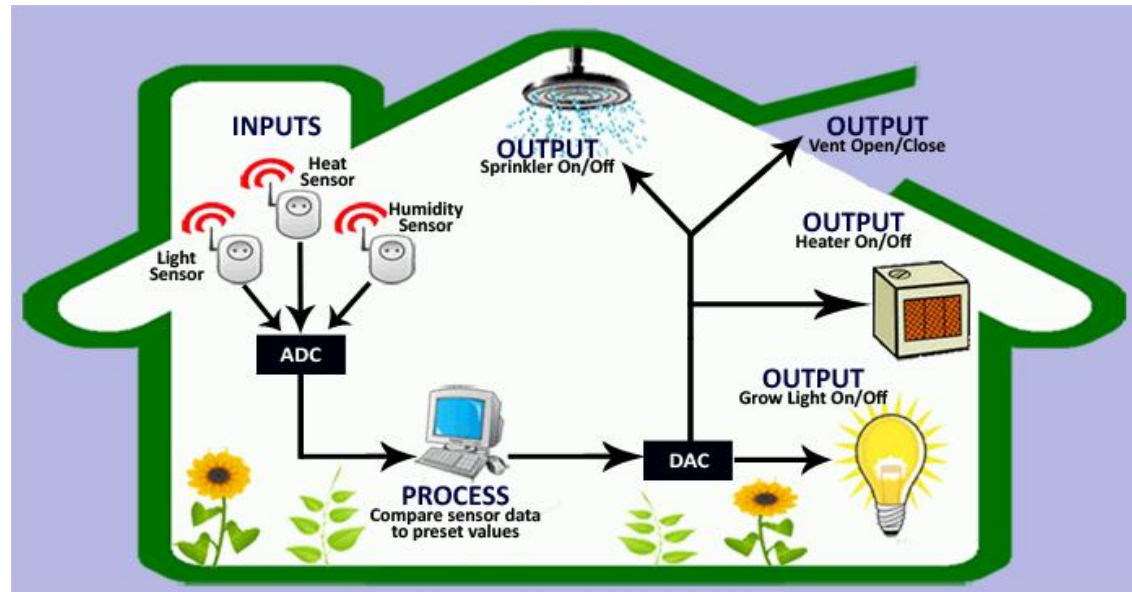
Control applications (devices) needed:



Computer controlled greenhouses **automatically detect humidity levels** and switch sprinklers on when

- **Grow lights to make plants flourish**
- **Motor to turn sprinkler on if plants need water**
- **Heater to warm the greenhouse**
- **Motor to open window vent to cool greenhouse down if it gets too hot.**

the soil becomes too dry



Click Image to zoom

How the system works:

- 1 Computer **reads data** from sensors and **compares** it to **preset values**.
- 2 If it is **too hot** = **heater turned off** and **vent opened**.
- 3 If it is **too cold** = **heater switched on** and **vent is closed**.
- 4 If there is **enough light** = **grow light switched off**.
- 5 If it is **too dark** = **grow light switched on**.
- 6 **Wet soil** = **sprinkler switched off**.
- 7 **Dry soil** = **sprinkler switched on**.

NOTE!

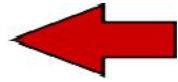
The process will **repeat constantly**. Sensors continually feed new data into the system and the computer will make decisions based on that data.

5 Other Examples.....

Other examples of devices that can be controlled by a computer include:

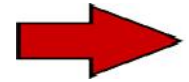
- **Automatic Cookers**
- **Automatic Washing Machines**
- **Microwaves Ovens.**

These devices were discussed in **section 6.2: [The effects of ICT on society](#)**



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